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A new method of screening for diabetic neuropathy using laser Doppler and photoplethysmography

Sung Woo Kim · Soo Chan Kim · Ki Chang Nam ·
Eun Seok Kang · Jae Jung Im · Deok Won Kim

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Abstract The purpose of this study is to suggest a simple, new method of screening for diabetic neuropathy. We measured blood volume changes by photoplethysmography (PPG) and blood perfusion by laser Doppler (LD) in the index fingers and big toes in 40 control subjects and in 50 (19 mild, 17 moderate, and 14 severe based on the nerve conduction velocity (NCV) test) and 35 diabetic patients with and without neuropathy, respectively. According to the results of PPG and LD measurements, the toe to finger ratios obtained from the neuropathic group were significantly higher than those from the control ($p < 0.001$) and the non-neuropathic groups ($p < 0.001$). Based on the NCV, the sensitivity of the LD method (92.0%) was higher

than that of the PPG method (84.0%) for both left and right sides. Although specificity of the LD (92.8%) was also higher than the PPG (84.3%) bilaterally, the PPG showed better reproducibility (5.5 versus 9.5%) and a significant ratio increase with severity, while the LD did not. Our suggested PPG method using the toe to finger ratio is reliable, simple, economical, and accurate, and could become an effective new screening tool for the early detection of diabetic neuropathy.

Keywords Diabetic neuropathy ·
Photoplethysmography · Laser Doppler ·
Blood volume change · Toe to finger ratio

S. W. Kim · D. W. Kim (✉)
Department of Medical Engineering, College of Medicine,
Yonsei University, Seoul 120-752, South Korea
e-mail: kdw@yuhs.ac
URL: <http://ybil.yonsei.ac.kr/>

S. C. Kim
Graduate School of Bio and Information Technology,
Hankyong National University, Anseong, South Korea

K. C. Nam
National Institute of Advanced Industrial Science
and Technology (AIST), Tsukuba, Japan

E. S. Kang
Division of Endocrinology and Metabolism,
Department of Internal Medicine, College of Medicine,
Yonsei University, Seoul, South Korea

J. J. Im
Division of Electronics and Information Engineering,
Chonbuk National University, Chonju, South Korea

1 Introduction

The incidence of diabetes is increasing dramatically due to economic development and lifestyle changes. The Growth from Knowledge (GfK) market measures (US Diabetes Patient Market Study, 2005) reported that the diabetic population in the United States had increased by approximately 86% over the past decade. The Korean Diabetes Association (Ministry of Health and Welfare, Korean Institute of Health and Society, 2004) reported that the Korean diabetic population increased approximately ten-fold from 1970 to 2001. Among many other diabetic complications, diabetic foot disease is considered one of the most serious as it may cause ulceration and subsequent amputation of the legs. It was reported that approximately 30,000 diabetic patients undergo foot surgery each year in the United States [5]. Amputations, however, are preventable by early diagnosis.

The causes of diabetic foot disease include neuropathy, neuro-ischemia, and ischemia. Of these etiologies,

neuropathy is the most common and accounts for up to 80–90% of the diabetic foot population [19]. Diabetic neuropathy results from nerve damage and can lead to loss of sensation or to abnormal feelings in the feet. It may even increase the likelihood of foot injuries developing into ulcers [5]. The nerve conduction velocity (NCV) test has been considered the gold standard method for diagnosing diabetic neuropathy, but it requires the application of a strong electrical stimulus to nerves, causing discomfort and pain in patients [17].

Many studies have been conducted by measuring blood volume changes in the fingers and toes because it was reported that patients with diabetic neuropathy have increased blood flow to their toes [6, 8–10, 16, 21]. Decreased sympathetic tone in the feet of neuropathy patients results in an open arteriovenous shunt (AVS). This causes increased blood flow in the feet [4] and toes, but does not have an effect on blood flow in the fingers [6]. On the contrary, diabetics with neuropathy showed smaller finger pulp and larger toe pulp blood flows than non-diabetics using laser Doppler (LD) [20]. Therefore, the toe to finger blood flow ratio in diabetics with neuropathy is larger than that in diabetics without it due to the presence of constant or decreased finger blood flow coupled with an increased toe blood flow. Based on these findings, this ratio may be a promising new screening parameter for diabetic neuropathy. In addition, by using the ratio, we can minimize variation due to difference in absolute blood volume changes among subjects and in skin temperature during measurement.

Laser Doppler has been used for measuring foot microcirculation [1, 11, 14], and photoplethysmography (PPG) has been used for monitoring blood perfusion in skin, venous reflux conditions, and skin flaps during plastic surgery [2]. In this study, the noninvasive techniques of PPG and LD were used to measure the blood volume changes and perfusions of the fingers and toes, respectively. We found optimal ratios for PPG and for LD that can distinguish diabetic patients with and without neuropathy. In addition, we determined the sensitivity and specificity using the NCV test, and the reproducibility of both PPG and LD.

2 Materials and methods

2.1 Subjects

Three groups of subjects were studied. The first group included 40 healthy, non-diabetic subjects, the second group included 35 diabetic patients without neuropathy, and the third group included 50 diabetic patients with neuropathy. Of those 50 diabetic patients with neuropathy,

there were 19 mild, 17 moderate, and 14 severe as determined by the NCV test. The diabetic patients with neuropathy had been diagnosed by the NCV test at the Yonsei University Medical Center, Seoul, Korea.

As shown in Table 1, all groups were matched for sex, age, and body mass index (BMI), and the two diabetic groups were also matched for type and duration of diabetes. The control subjects had a fasting glucose level between 3.9–5.8 mmol/L and less than 6.7 mmol/L 2 h after breakfast. None of the control subjects reported current use of any medications. All subjects were informed about the purpose of and procedure for the study and subsequently gave their informed consent. The Yonsei Medical Research Ethics Committee reviewed the full protocol and approved this study.

2.2 Hardware for PPG measurement system

Our constructed PPG measurement system utilized four channels to allow the simultaneous measurement of the PPG signals from the left and right fingers and toes. The wavelengths of the PPG sensors (DS0-100A Durasensor, Nellcor, USA) were 660 nm (red) and 940 nm (infrared). Signal amplification, filtering, and normalization were performed using the operational amplifiers (TL082, Texas Instruments, USA), 10 Hz low pass filter, and PIC microcontroller (PIC 16C711, Microchip, USA), respectively. The final signal was sent to a notebook computer (Sens V20, Samsung, Korea) through a DAQ-pad (PCI-6020E,

Table 1 Characteristics of experimental groups

	Control	Diabetic groups	
		Non-neuropathy	Neuropathy
Number of subjects	40	35	50
Sex (M/F)	18/22	15/20	21/29
Age (years)	65.8 ± 8.9	61.0 ± 8.0	65.1 ± 9.0
Body mass index (kg/m ²)	22.6 ± 1.4	23.5 ± 2.7	22.9 ± 3.3
Systole (mmHg)	121.2 ± 8.5	126.5 ± 15.0	131.1 ± 18.9
Diastole (mmHg)	79.5 ± 3.2	78.8 ± 10.2	77.9 ± 10.4
Diabetes duration (years)	–	16.7 ± 6.7	13.7 ± 8.6
Type I	–	1	2
Type II	–	34	48
Smokers	9	7	4
HbA _{1c} (%)	–	10.2 ± 5.6	9.8 ± 2.8
Fasting glucose (mmol/L)	<5.8	8.3 ± 2.1	8.8 ± 4.0
HDL (mg/dL)	–	52.9 ± 15.1	50.2 ± 17.6
LDL (mg/dL)	–	122.1 ± 13.9	102.7 ± 35.2

National Instruments, USA). For accurate and precise measurement, the PPG signal measurement system was calibrated by connecting the output of the SpO₂ simulator (Index[®]2_{XLFE}, Fluke, USA) to the input of the PPG system. Then, the output signals from both instruments were matched by adjusting the potentiometer of the PPG system for all four channels.

2.3 Software of PPG signal measurement system

LabVIEW 6.1 (National Instrument, USA) was used to develop the real time data acquisition and signal analysis program. The amplitudes representing the differences between the peaks and valleys of each red LED waveform were averaged within the selected window to obtain the mean blood volume changes of the fingers and toes. They were then used to obtain the left and right blood volume change ratios. Finally, the toe to finger ratio was used to minimize the difference in the absolute blood volume and skin temperature between each subject.

2.4 Experimental procedure

2.4.1 Nerve conduction velocity test

The NCV test (Neuroscreen, Jaeger and Toennies, Wuerzburg, Germany) was performed by a clinician at the Yonsei University Medical Center, Seoul, Korea. An active electrode was placed over the nerve segment being studied. The median and ulnar nerves were tested for the upper limb, and the peroneal, tibial, and sural nerves were tested for the lower limb. The motor nerves (median, ulnar, peroneal, and tibial nerves) and sensory nerves (median, ulnar, and sural nerves) were examined to determine the presence of neuropathy. Diabetic patients participating in this study were diagnosed with mild, moderate, or severe neuropathy if abnormality occurred on 1–2 nerves, 3–5 nerves, or 6–7 nerves, respectively. A total of 50 diabetic patients were diagnosed with neuropathy by the clinician.

2.4.2 Measurement procedure

The procedure for the measurements was as follows. Subjects rested in the supine position for a minimum of 10 min before beginning the experiment. The PPG signals from the index finger and first toe for both the left and right sides were simultaneously recorded in triplicate in the supine position by our constructed system. Bilateral blood perfusion and the temperatures of fingers and toes were

also simultaneously recorded in triplicate using an LD perfusion monitoring and temperature unit (PF 5010 and 5020, Perimed, Sweden). The small angled thermostatic probes (457, Perimed, Sweden) were used to measure perfusion and temperature simultaneously with double-sided adhesive strips (PF 10–3, Perimed). PeriSoft for Windows (ver 2.5, Perimed) software was used for data storage and analysis. Each measurement lasted 30 s and was recorded three times in order to verify the reproducibility of PPG and LD. The electrodes were replaced three times for each repeated PPG and LD measurement. For later analysis, stable 10-s intervals of PPG and LD signals were selected. The room temperature was maintained at 23°C during the experiment.

2.5 Statistical analysis

All data are shown as means and standard deviations. $p < 0.05$ was considered statistically significant. The independent sample *t*-test and one-way ANOVA test were performed using SPSS 10.0 for Windows (SPSS Inc, Chicago, IL, USA). The Bonferroni multiple comparison method was performed for further analysis in the ANOVA test.

3 Results

3.1 Clinical characteristics

There were no statistically significant differences among the three groups in age ($p = 0.099$), BMI ($p = 0.261$), systolic ($p = 0.236$) or diastolic ($p = 0.680$) blood pressures (Table 1). There were also no statistically significant differences between the two diabetic groups in diabetes duration ($p = 0.380$), glycolysed hemoglobin A_{1c} (HbA_{1c}, $p = 0.493$), fasting glucose ($p = 0.429$), high density lipoprotein (HDL, $p = 0.466$), or low density lipoprotein (LDL, $p = 0.053$).

3.2 Blood volume change and skin temperature

Because there were no statistically significant differences between the left and right sides of fingers and toes in blood volume changes and perfusion, the left and right blood volume changes and perfusion were pooled. Blood volume changes and temperatures of fingers and toes were obtained by the PPG for the control subjects and patients with and without neuropathy. The toe temperature, ranging from 25 to 34°C, had a wider distribution than that of the finger which ranged from 29 to 35°C. There were no significant

differences in the toe and finger temperatures among the three groups.

Figure 1 shows finger and toe blood volume changes measured by the PPG (a, b) and LD (c, d) for the five groups. The bold lines inside the box plots indicate median values. The upper and lower lines of the box are at the 25th and 75th percentiles and the top and bottom whiskers are the highest and lowest values, respectively. The significance p levels were shown among the control, non-neuropathic, and neuropathic groups pooling the three groups, and among the mild, moderate, and severe groups. As expected in Fig. 1a, b, the PPG method showed no significant differences in blood volume changes between the control and non-neuropathic groups in the fingers and toes, respectively. However, there was a significant decrease in the blood volume changes of the neuropathic group compared with that of the non-neuropathic group in the fingers ($p < 0.01$) and a significant increase in the toes ($p < 0.001$). Furthermore, there was a significant decrease in the blood volume changes of the neuropathic group compared with that of the control group in the fingers ($p < 0.001$) and a significant increase in the toes

($p < 0.001$). The above finding is consistent with the study wherein 76% of diabetics with neuropathy showed smaller finger pulp and larger toe pulp blood flows than non-diabetics [20]. Therefore, this finding not only validated previous studies [6, 8–10, 13, 16, 20] but confirmed our assumption that the toe blood volume changes in the neuropathic group would be larger than those in the non-neuropathic group.

As shown in Fig. 1c and d, there was a significant difference in the finger blood perfusion measured by the LD between the control and non-neuropathic groups ($p < 0.01$). This finding by LD is different from that by PPG, and while future studies are required to further investigate this discrepancy, it may be due to the different methodology and measured volume sizes of the PPG and LD methods. Though the toe blood perfusion of the neuropathic group measured by LD was larger than those of the control and non-neuropathic groups, it was not significant ($p > 0.05$). This finding is consistent with a Nabuurs–Fransen study [14] wherein foot LD flux in patients with peripheral polyneuropathy was higher than that of the control without significance (7.4 versus 5.9). While the measured position

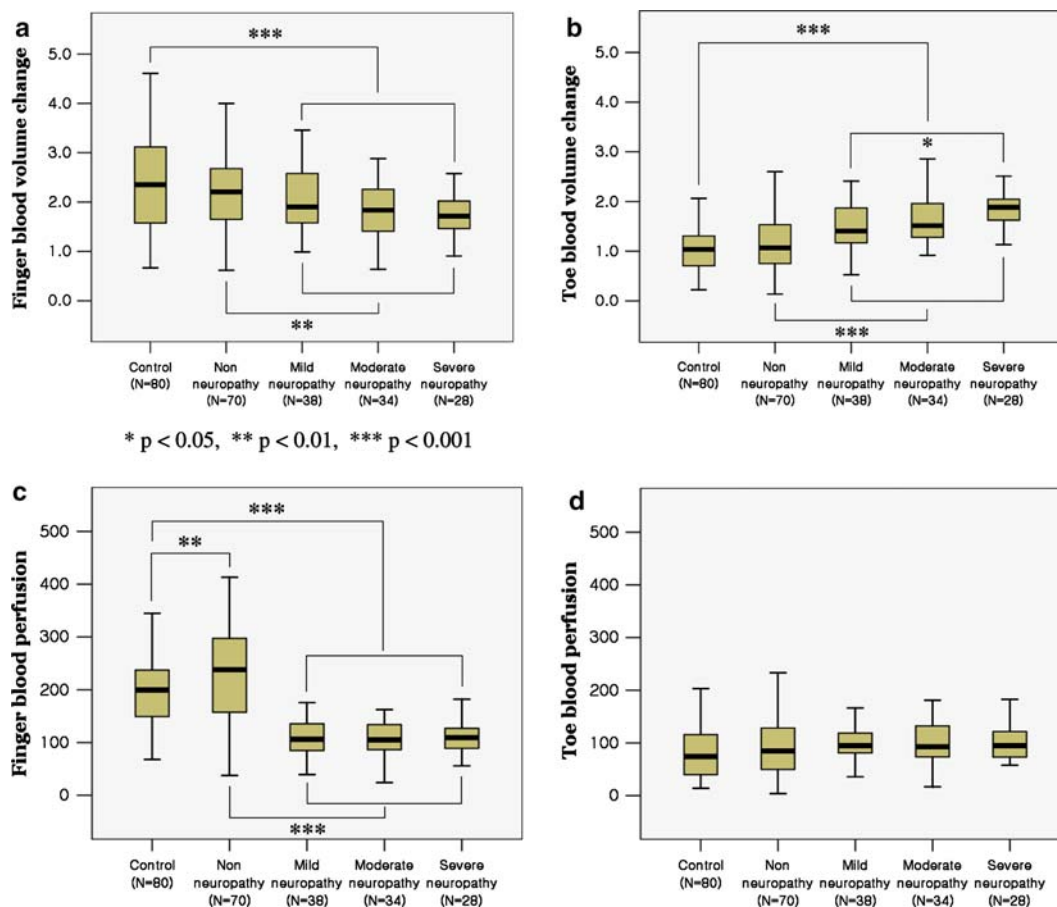


Fig. 1 Box plots for finger (a) and toe (b) blood volume changes by PPG, and for finger (c) and toe (d) blood perfusion for the five groups

(foot versus toe) was different between their study and ours, both studies showed the same trend.

The above disparity between the PPG and LD was mitigated or eliminated by utilizing the ratio of toe to finger blood volume change (or perfusion) as discussed in the following section (Fig. 3).

3.3 Blood volume change ratios and temperature differences

Figure 2 shows the temperature difference between fingers and toes and the ratios of blood volume change by PPG (a) and perfusion by LD (b) for the control ($n = 80$), non-neuropathic ($n = 70$), and neuropathic groups ($n = 100$). The temperature differences of the control and non-neuropathic groups were widely distributed in the range of 0–8°C, while those of the neuropathy group were distributed from 0 to 4.5°C. There were no neuropathy patients with a temperature difference greater than 4.5°C. This finding suggests that a subject is not considered neuropathic if his or her temperature difference is higher than 4.5°C. The two vertical lines in Fig. 2a, b are the optimal boundaries differentiating neuropathic from non-neuropathic and control groups, and will be explained later in this paper.

Figure 3 shows the box plots for the calculated ratios for the five groups by the PPG and LD methods. There were significant differences in the blood volume change ratios between the control and neuropathic groups ($p < 0.001$) as well as between the non-neuropathic and neuropathic groups ($p < 0.001$), but not between the control and non-neuropathic groups ($p > 0.05$) by both methods. The difference between the PPG and LD methods is that there are ratio increases with severity for PPG but not for LD for the neuropathic group. In this sense, PPG is a superior method.

The reproducibility (standard deviation/mean $\times 100\%$) of the PPG method for the control, non-neuropathic, and neuropathic groups was 8.0 ± 5.9 , 2.8 ± 1.9 , and $5.4 \pm 4.9\%$, respectively. The total mean reproducibility was $5.5 \pm 5.1\%$. Those of the LD were 13.8 ± 9.1 , 5.3 ± 3.7 , and $9.0 \pm 6.1\%$, respectively. The total mean reproducibility was $9.5 \pm 7.5\%$. Reproducibility is one of the most important factors for reliable diagnosis in medicine, and in this regard PPG is the superior method.

3.4 Sensitivity and specificity of PPG and LD

Sensitivity and specificity were calculated as (1) and (2) where TP = true positive, TN = true negative, FP = false positive, and FN = false negative.

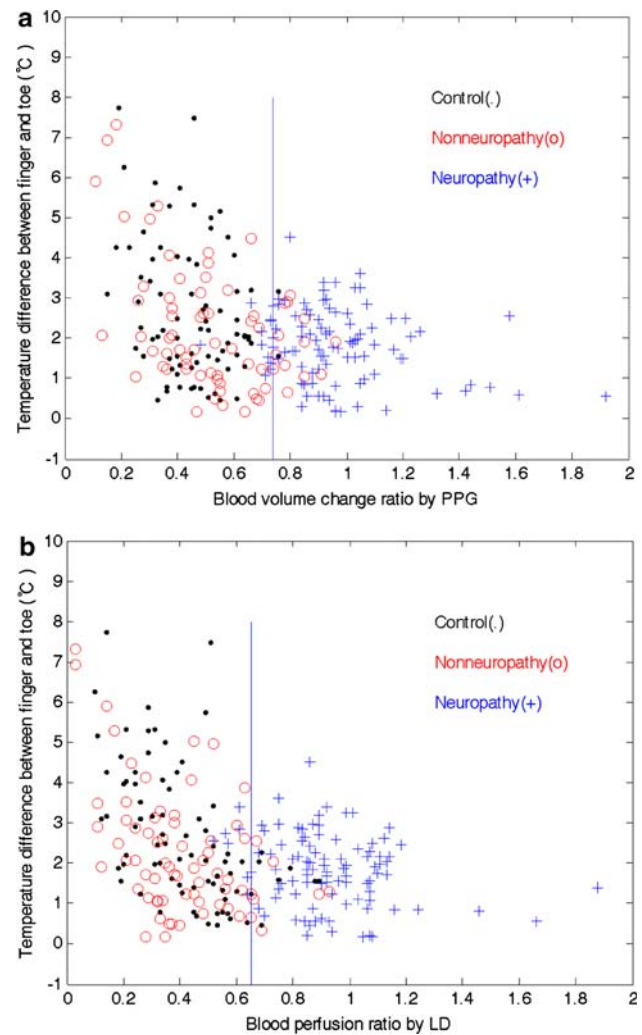


Fig. 2 PPG blood volume change ratio (a) and LD perfusion ratio (b) with temperature differences between fingers and toes by the LMS method ($n = 80$ for control, $n = 70$ for non-neuropathy, $n = 100$ for neuropathy)

$$\text{Sensitivity}(\%) = \frac{\text{TP}}{\text{TP} + \text{FN}} \times 100 \quad (1)$$

$$\text{Specificity}(\%) = \frac{\text{TN}}{\text{TN} + \text{FP}} \times 100 \quad (2)$$

We applied Bayesian, least mean square (LMS), and ROC curve methods to assess the sensitivities and specificities of PPG and LD, which account for the inherent trade-off between sensitivity and specificity of the tests by determining the appropriate boundary value for each method [7, 12, 15]. The vertical lines in Fig. 2a, b represent the optimal PPG and LD ratios of 0.74 and 0.65, respectively, for distinguishing neuropathic from non-neuropathic diabetes using LMS as shown in Table 2. If the vertical line in Fig. 2a moves to the right, the boundary value of the blood volume change ratio

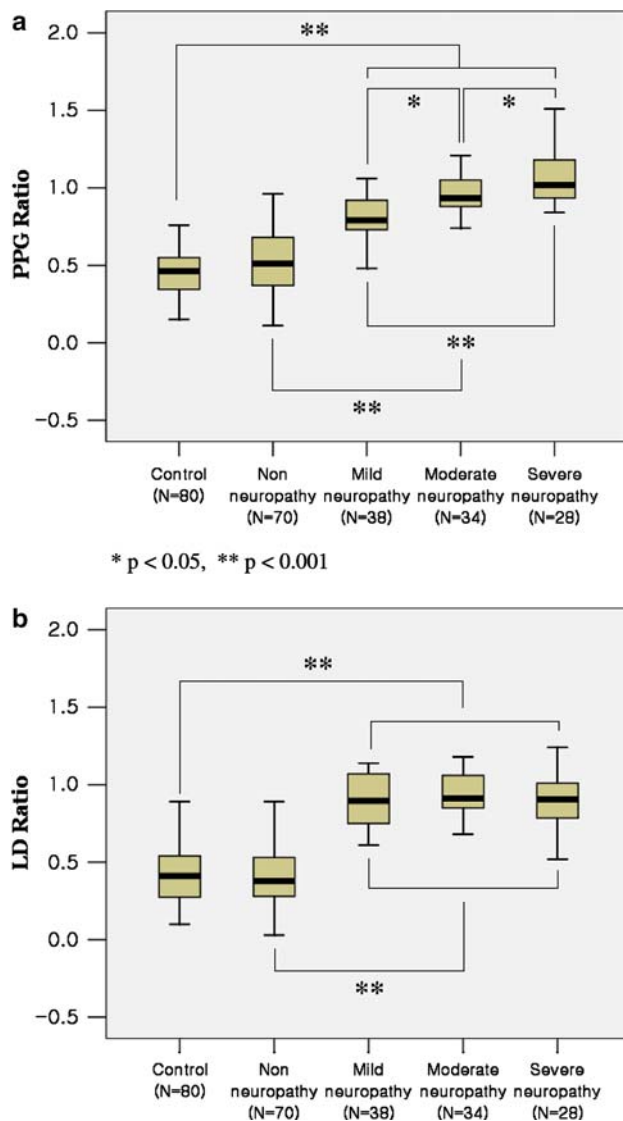


Fig. 3 Box plots for PPG blood volume change ratio (a) and LD perfusion ratio (b) for the five groups

increases. At the same time, the sensitivity decreases and the specificity increases since the blood volume change ratio of neuropathic diabetes is larger than that of non-neuropathic diabetes.

Table 2 shows the calculated sensitivities and specificities with the corresponding optimal ratios of PPG and LD methods for the Bayesian, LMS, and ROC curve methods. The LD method showed better sensitivity and specificity than the PPG for all three methods. In clinical applications, the higher the sensitivity of a test, the better the diagnosis. Therefore, the lowest boundary value of 0.65 is considered optimal for LD, and thus the corresponding sensitivity and specificity are 93.0 and 91.4%. The PPG method by LMS has a sensitivity of 86.0% and specificity of 82.8%. While different tests for neuropathy have low correlations among themselves [13, 18], both of our proposed methods

Table 2 Sensitivity, specificity, and boundary values calculated by the Bayesian, LMS, and ROC methods for both PPG and LD

Method	Sensitivity (%)	Specificity (%)	Boundary value
PPG			
Bayesian	79.0	89.0	0.79
LMS	86.0	82.8	0.74
ROC	84.0	84.3	0.75
LD			
Bayesian	81.0	97.1	0.75
LMS	93.0	91.4	0.65
ROC	92.0	92.8	0.66

demonstrated satisfactory sensitivities and specificities. This may be due to normalization with the toe to finger ratio, which minimizes the variation of absolute blood flow between subjects and the influence of skin temperature.

4 Discussion and conclusion

The toe to finger ratios of the neuropathic group by LD was increased by a decrease in finger blood perfusion but not by an increase in toe blood perfusion as shown in Fig. 1c, d. When we compared our results to those from a previous study (<http://www.your-feet.com/pages/diabetic.aspx>), the toe blood flow by LD in diabetic neuropathic patients did not differ significantly from blood flow in controls and non-neuropathic diabetic patients. This result is in accordance with our measured toe blood flow using LD. However, other studies have shown conflicting results. In these studies, the feet of diabetic patients with neuropathy showed increased skin blood flow when compared with those of diabetic patients without neuropathy and control subjects [4]. These results are in accordance with our results from measuring toe blood flow using PPG.

Wigington showed that finger blood flow in patients with diabetes was lower than in those without it [20], a result similar to ours using PPG. Though our proposed ratio method cannot discriminate between finger and toe neuropathy, it has the potential to become an effective, new screening tool for the detection of diabetic neuropathy as it most commonly affects feet before hands [3, 20].

Another advantage to our proposed method is that toe to finger ratios with a diabetic neuropathic foot would be increased either by decreased finger blood volume coupled with increased toe blood volume as seen with our PPG method, or by decreased finger blood perfusion with unchanged toe blood perfusion like that seen with our LD.

The LD and PPG methods are different both in principle and in region of measurement. The LD used in this study is a reflected mode and the distance between the transmitting and receiving fibers is only 0.25 mm, and thus the

measuring volume or depth (0.5–1 mm) is very small. Conversely, PPG is a transmitted mode and the distance between the transmitting and receiving transducers is the depth of the finger or toe being measured, and thus its measuring volume is considerably larger than that of LD. Therefore, both values cannot and should not be the same, although they may show the same trend in part, as evidenced by our results. In this study, we used LD in order to indirectly support the validity of the PPG measurements because LD has previously been used in many studies.

One of the most important findings in this study is that the blood volume change ratio of toe to finger may distinguish neuropathic diabetes from non-neuropathic diabetes with a high sensitivity and specificity. The suggested PPG method using this ratio provided a sensitivity of 86.0%, a specificity of 82.8%, and a mean reproducibility of 5.5%, while the LD showed a higher sensitivity of 93.0% and a higher specificity of 91.4%, but a lower mean reproducibility of 9.5%. While the LD method is superior in its sensitivity and specificity, it is expensive, complex, and has relatively poor reproducibility compared with that of PPG. The PPG method also showed proportionally increased ratios with neuropathic severity while the LD did not. The suggested PPG system has proven to be highly reproducible, simple, economical, and accurate and has opened up the possibility for its use as an effective new screening tool for the early detection of diabetic neuropathy.

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